

Multilingual and Retrieval-Augmented Temporal Reasoning with TISER

Kerem Kose (s339018), Arian Mohammadi (s346278), Sheharyar Zafar (s346351)

Deep Natural Language Processing

Politecnico di Torino

Turin, Italy

Abstract—TISER (Bazaga et al., 2025) improves temporal reasoning by making a model emit an explicit trace—step-by-step reasoning, a constructed timeline, and a self-reflection step—before its final answer. The original work targets 7B English-only models. We reproduce the TISER format on a 3B model (Qwen2.5-3B-Instruct) fine-tuned with QLoRA (4-bit NF4) on a single 8GB GPU, and extend it in two directions. (1) **Multilingual**: an NLLB-200 pipeline with entity caching, proper-noun shortcutting, hallucination detection, and template localization produces structurally faithful Italian, German, and French training data. (2) **RAG**: as a controlled ablation, we index training questions with a multilingual sentence encoder (FAISS) and inject the most similar solved example as a few-shot demonstration, asking whether retrieval helps a benchmark whose samples are *already self-contained*. On English, retrieval and fine-tuning act as *substitutes*: few-shot retrieval lifts the untuned base model (F1 0.69 $\rightarrow$ 0.74, $N=100$) but slightly hurts the fine-tuned model (0.92 $\rightarrow$ 0.90); on a mixed four-language set retrieval is no longer harmful for the fine-tuned model and trends positive (+0.03 F1, directional at $N=100$). Fine-tuning lifts English F1 from 0.690 to 0.912; a single four-language fine-tune matches this on English (0.909) while adding IT/DE/FR at F1 0.865–0.890 with English leakage below 0.01, whereas an English-only fine-tune transfers only partially.

Index Terms—temporal reasoning, large language models, QLoRA, retrieval-augmented generation, multilingual NLP, timeline self-reflection

I. INTRODUCTION

Task. Given a natural-language *temporal question* and a *temporal context* (a set of dated facts or events), a model must produce the correct answer. Formally, the input is a triple (C, Q, L) where C is a self-contained set of temporal facts, Q is a temporal query (entity-at-time, ordering, duration, boolean, or “unknown”), and L is the language. The expected output is a structured trace—a `<reasoning>` block containing a `<timeline>` and a `<reflection>`—followed by a final `<answer>`. Evaluation targets only the `<answer>` payload; the trace is the mechanism, not the scored object. We use the five TISER categories: `tgqa`, `tempreason_l2`, `tempreason_l3`, `timeqa_easy`, and `timeqa_hard`. `tgqa` answers are free-form event descriptions (hardest); the others are largely entity/date/boolean answers.

Contributions.

- 1) *Small-model TISER*: the full timeline-self-reflection pipeline on a 3B model with QLoRA on an 8GB GPU.
- 2) *Multilingual TISER* via an NLLB-200 translation and quality-control pipeline (IT/DE/FR), yielding one fine-

tuned model that preserves English while adding three languages at ≤ 4 F1 spread—plus a cross-lingual baseline showing an English-only fine-tune leaves 7–11 F1 on the table.

- 3) *A RAG ablation* isolating *when* retrieval helps temporal reasoning: substitutes with fine-tuning on English, small complementary value where fine-tuning is weakest.

Source, data pointers, and run scripts: <https://github.com/arianmo477/tiser-multilingual-rag>.

II. RELATED WORK

TISER [1] introduces timeline self-reflection: the model constructs an explicit timeline and critiques it before answering, improving temporal consistency. **QLoRA** [2] enables fine-tuning of quantized (4-bit NF4) models with low-rank adapters, the key that lets a 3B model train on 8GB. **NLLB-200** [3] is the multilingual translation backbone for our IT/DE/FR data. Dense retrieval builds on **Sentence Transformers** [4] embeddings and the **FAISS** [5] similarity index. **RAG** [6] augments generation with retrieved passages; it was designed for *open-domain* knowledge gaps—a regime self-contained TISER samples lack, motivating our ablation. The source QA data derive from **TimeQA** [7], **TempReason** [8], and **TGQA** [9].

III. DATA AND PREPROCESSING

A. Normalization and validation

The raw TISER releases are JSON lists whose samples carry a composite prompt, the reference output trace, an answer, a `question_id`, and a `dataset_name`. A validation pass runs before any training or translation: broken unicode escapes (a literal `uXXXX` with the backslash stripped) are detected and re-decoded, without which accented entity names are corrupted and exact match silently fails; the `question` and `temporal_context` fields are recovered from the composite prompt via the fixed markers `Question:` and `Temporal context:`; duplicates are removed by `question_id`; and missing-information answers (“unknown”, “not specified”, ...) are kept as legitimate TISER labels but counted, with optional flags to remove them for controlled experiments. Rejected samples are exported to an `invalid_samples` file rather than silently dropped, and

the validator emits a statistics JSON (per-language and per-error counts plus a SHA-256 of the output) so every result is traceable to the exact data artifact that produced it.

B. Frozen snapshot, sampling, and mixing

During development we found a bug in the class-balanced subsampler (its top-up step could re-select an already-chosen sample). Fixing it changes which 15 000 English samples are drawn, which would silently invalidate previously reported numbers; we therefore froze the exact 15k subset as an artifact (`train_tiser_15000_en.json`) and added a provenance script that re-derives it from the full split by re-executing the historical pipeline—the same `shuffle(seed=42)`, PASS filter, and the *pre-fix* sampler embedded verbatim—and asserts equality on `question_id`. New experiments use the corrected sampler. Two deterministic samplers (seed 42) drive all subsampling: one balances across `dataset_name` with equal quotas and duplicate-free top-up; the other buckets by the (*language, dataset*) pair, so a mixed $N=500$ evaluation contains exactly 25 samples per cell. Multilingual training files are built by a mixer that stamps each sample’s language (inferred from the file path, restricted to `en/it/de/fr`) and applies the same exact per-cell balance. All non-English data additionally pass the quality gate of Section IV-D: only `validation_status = PASS` samples are used for training and evaluation.

IV. METHODOLOGY

A. Fine-tuning pipeline

The base model is Qwen/Qwen2.5-3B-Instruct [10] in 4-bit NF4 with double quantization; LoRA [11] adapters cover every attention and MLP projection, discovered by module-name inspection. Supervision is completion-only: the per-language instruction, context, and question form the prompt, whose token positions are masked to -100 ; the loss is computed only on the reference trace plus EOS. Sequences are truncated at 1536 tokens and samples whose target is entirely truncated away are dropped; a custom collator pads labels with -100 to keep the mask aligned. A 10% validation split (filtered for degenerate label-less samples) selects the best checkpoint by validation loss, with checkpointing tied to the evaluation interval. The 8 GB budget is met with fp16 compute, paged 8-bit AdamW, gradient checkpointing, and an explicit 7 GiB-GPU/CPU-offload memory map; Table I lists the configuration.

B. Inference

Prompts reuse the training templates (`tiser_full_{en, it, de, fr}.txt`, English fallback) and the `<reasoning>` cue is prepended to the decoded continuation so the parsed trace is always well-formed. Decoding is greedy (`repetition_penalty= 1.1`, left padding) with a custom stopping criterion that decodes the last 40 generated tokens after each step and halts at `</answer>`. Generation is *iterative*: if `</answer>` is missing after the initial 768-token budget, the partial output is fed back

TABLE I
TRAINING CONFIGURATION (IDENTICAL FOR ALL MODELS).

Base model	Qwen2.5-3B-Instruct, 4-bit NF4
LoRA	$r=16$, $\alpha=32$, dropout 0.05
Max sequence length	1536 tokens
Batch / grad. accum.	1 / 16 (effective 16)
Optimizer / LR	paged AdamW 8-bit / 3×10^{-4}
Epochs / val. split	2 / 10%
Precision / memory	fp16, grad. ckpt., 7 GiB GPU + CPU offload

and extended in 256-token steps (up to twice), recovering the long traces free-form `tgqa` answers occasionally need. The answer is extracted from the `<answer>` payload with tolerant fallbacks for minor tag corruption, then normalized before scoring. Appendix ?? sketches the retrieval path in pseudocode.

C. Multilingual extension

NLLB-200-distilled [3] translates the English data to Italian, German, and French. Applied naively it destroys structure: entity names drift between occurrences (breaking exact match on correct answers), year markers get rewritten as prose, and short entity strings get expanded into descriptive sentences (“Penn Central” $\rightarrow$ “Penn Central est une ville d’Angleterre”). Structurally, the translator never sees what must not change: text is split on the TISER tags so they pass through verbatim, and parenthesized event mentions are masked with placeholders before the NLLB call and restored after, preserving numbering and year markers. Three quality layers target the remaining failure modes.

(a) Entity cache. Every parenthesized entity—in question, context, timeline, and answer—is translated once, cached, and reused at every occurrence, enforcing a single surface form per entity (otherwise the same club can appear as “Herz von Midlothian” and “Herz von Mittellothian” in one sample). The cache is persisted and shared across runs; outputs are written atomically, making multi-day jobs resumable.

(b) Proper-noun shortcut and hallucination guard. Person and place names of one to three capitalized words bypass NLLB entirely—they translate to themselves, yet NLLB often hallucinates on them. As a fallback for other short inputs (<60 chars), outputs that grew more than $2\times$ and match descriptive patterns (“is a city”, “*est une*”) are rejected in favor of the source string, and *carrier prefixes* NLLB sometimes emits (“*Übersetzen Sie diesen Text:*”) are stripped.

(c) Template localization. High-frequency scaffolding present in every sample—month names, boolean labels (*Vero/Falso, Wahr/Falsch, Vrai/Faux*), “starts at / ends at”, possessive templates, and the most common question stems—is translated by hand-written per-language rules: NLLB renders these phrases consistently *within* a call but inconsistently *across* calls, which corrupts training signal. The same cache and defenses also translate the `output` field (the full reasoning trace) in the training set.

TABLE II

MODEL $\times$ TEST-SET OVERVIEW. BASE AND EN-ONLY ROWS USE 500 BALANCED HELD-OUT SAMPLES PER TEST SET; THE EN+IT+DE+FR ROWS COME FROM ONE MIXED 500-SAMPLE EVALUATION (4-LANG AVG), WHOSE PER-LANGUAGE ROWS ARE ITS 125-SAMPLE SLICES. $\Delta F1$ IS THE MIXED MODEL’S F1 ADVANTAGE ON THE SAME TEST LANGUAGE.

Model	Test set	F1	chrF	NormEM	EM	SoftEM	EngLeak	$\Delta F1$ (Mixed)
Base (no fine-tune)	EN	0.690	69.88	0.580	0.574	0.648	0.000	—
EN only (15k)	EN	0.912	89.38	0.840	0.836	0.904	0.000	−0.003
	IT	0.808	82.42	0.678	0.674	0.748	0.062	+0.078
	FR	0.795	80.66	0.658	0.652	0.714	0.058	+0.070
	DE	0.783	79.33	0.674	0.672	0.758	0.048	+0.107
EN+IT+DE+FR (15k, Mixed)	4-lang avg	0.887	88.42	0.794	0.788	0.854	0.006	—
	EN	0.909	90.27	0.824	0.824	0.896	0.000	—
	IT	0.886	89.50	0.800	0.784	0.848	0.008	—
	FR	0.865	86.02	0.760	0.752	0.808	0.008	—
	DE	0.890	87.87	0.792	0.792	0.864	0.008	—

D. Translation quality control

A scorer computes cosine similarity between each translated field (question, temporal_context, answer, output) and its English source with a multilingual sentence encoder, batching all pairs in one pass. Hard structural checks additionally flag missing years, unresolved “()” artifacts, broken TISER tags, absence of the translated answer inside the translated output, and English leftovers (“Which event”, “starts at”). Numeric and boolean answers skip the similarity test and are judged structurally. A sample scores as its *worst* field; samples below threshold or with any hard error go to a failed set for retranslation, the rest form the passed set used throughout.

E. RAG extension

Rather than appending retrieved *evidence* (classic RAG), we retrieve a *similar solved training example* and inject it as a demonstration, because every target already carries its own complete context. Index construction: load a training split, drop corrupt samples (empty output, () artifacts), encode each sample’s *question text* with paraphrase-multilingual-MiniLM-L12-v2, L2-normalize, and store in a FAISS IndexFlatIP (cosine on normalized vectors) alongside the full sample dicts. At inference we embed the target question, cosine-search, drop hits below `min_score` (0.60), drop self-matches by `question_id`, and drop cross-language hits on a mixed index. Two modes: *few_shot* (default) prepends the retrieved question+context+solved output as a demonstration; *context_stuffing* appends the retrieved context to the target’s own. Retrieval matches by question *structure*, not entity identity, which is why it supplies format rather than facts (Appendix ??).

Experiment 1 (key study). We run four cells on identical samples (same test file, seed 42): `base_norag`, `base_rag`, `ft_norag`, `ft_rag`, and aggregate them aligned on shared `question_id`, reporting the RAG delta per model. The full 2 $\times$ 2 runs twice: on an English and on a mixed four-language test set. *Hypothesis*: RAG helps base (it supplies the

reasoning format the base model never learned) but is $\approx$ neutral or negative on `ft`, so retrieval and fine-tuning are substitutes.

V. EXPERIMENTAL SETUP

Hardware. A single 8GB consumer GPU (also validated on a Colab Tesla T4). **Software.** Python 3.10, PyTorch [15] 2.x, Transformers [14], PEFT, bitsandbytes, datasets, sentence-transformers, FAISS. **Metrics** (computed on the extracted, normalized `<answer>` payload): *EM* (strict normalized match); *NormEM* (sorted token bags, crediting word-order variants); *SoftEM* (substring containment either direction); *token-F1* (SQuAD-style [13]); *chrF* [12] (character *n*-gram F, robust to IT/DE/FR morphology); and *EngLeak*, the fraction of non-English samples whose prediction matches the *English* gold instead of the target-language gold—the model reasoned correctly but answered in the wrong language.

VI. RESULTS

A. Main models (multilingual)

Table II reports the models on 500 balanced held-out English samples. Fine-tuning is the dominant factor: it lifts the base model from F1 0.690 to 0.912 (+0.22), with the largest gains on `tempreason_l2` (0.517 $\rightarrow$ 0.830) and `tgqa` (0.515 $\rightarrow$ 0.829). The four-language model matches the English-only model on English within 0.004 F1, i.e. multilingual training induces no negative interference. Per-language F1 for the multilingual model is EN 0.909, IT 0.886, DE 0.890, FR 0.865—a 2–4 F1 spread, with French weakest due to residual NLLB noise on isolated entities. EngLeak stays ≤ 0.008 across all four languages.

B. Cross-lingual transfer of the English-only model

Is multilingual fine-tuning actually necessary? The cross-lingual rows of Table II put the EN-only model side by side with the four-language model on the same IT/DE/FR test sets (500 balanced samples each). Transfer is real but partial: the EN-only model lands at F1 0.78–0.81, i.e. 7.0–10.7 F1 *below* the multilingual model on every language. Equally telling is *how* it fails: EngLeak rises to 0.048–0.062 and concentrates in

the tempreason categories (up to 0.14), meaning the EN-only model often reasons correctly but answers in English—invisible to SoftEM yet fatal for a deployed multilingual system. The multilingual fine-tune eliminates this failure mode ($\text{EngLeak} \leq 0.008$) while closing the F1 gap, confirming the non-English gains come from multilingual training data, not the base model’s latent multilinguality.

C. RAG ablation (Experiment 1)

Table III reports the four cells on $N=100$ English and $N=100$ mixed samples (20 per category; the mixed set has 25 per language; $\text{few_shot}, \text{top_k}=1, \text{min_score}=0.60$). Hit-rate was 100% on English and 83% on mixed, where same-language exemplars are enforced and cross-language hits filtered out.

TABLE III
RAG ABLATION: BASE VS. FINE-TUNED, RAG ON/OFF ($N=100$ EACH).

Test	Cell	F1	chrF	NormEM	EM	SoftEM
EN	base_norag	0.6879	69.11	0.60	0.59	0.65
EN	base_rag	0.7369	73.15	0.61	0.60	0.66
EN	ft_norag	0.9225	89.99	0.86	0.84	0.92
EN	ft_rag	0.8989	87.57	0.82	0.81	0.89
Mixed	base_norag	0.5694	60.73	0.42	0.42	0.53
Mixed	base_rag	0.5623	60.82	0.37	0.37	0.57
Mixed	ft_norag	0.8498	84.08	0.74	0.74	0.80
Mixed	ft_rag	0.8818	86.89	0.78	0.77	0.84

English: substitutes, as hypothesized. The RAG effect (rag-norag) on the base model is F1 +0.049; on the fine-tuned model, F1 −0.024, EM −0.03. Few-shot retrieval lifts the un-tuned model—consistent with retrieval supplying the reasoning *format* the base model lacks—while for the fine-tuned model the exemplar adds no missing information and its extra context tokens slightly hurt. Per category (base, F1) the lift concentrates where formatting matters: tempreason_{13} $0.638 \rightarrow 0.770$, 12 $0.582 \rightarrow 0.674$, tgqa $0.543 \rightarrow 0.650$ in F1 while EM *drops* ($0.30 \rightarrow 0.25$)—retrieval nudges free-form answers closer in surface form without making them exactly right. On timeqa_hard the base model gets *worse* ($0.767 \rightarrow 0.691$): the exemplar can also distract where the model was already competent.

Mixed: a partial reversal. On the four-language set the base model no longer benefits (F1 −0.007, EM −0.05)—it struggles with non-English inputs regardless of the exemplar—while the fine-tuned multilingual model now *gains* F1 +0.032, EM +0.03, concentrated exactly where it has the most headroom: tempreason_{12} $0.744 \rightarrow 0.854$, timeqa_hard $0.922 \rightarrow 0.990$, 13 $0.875 \rightarrow 0.925$, against small losses on the already-strong timeqa_easy and tgqa . A same-language solved exemplar appears to anchor target-language phrasing where the noisier translated training data left residual weakness. Since decoding is greedy, the 17% of samples without an exemplar are identical to the no-RAG cell, so the gain is carried entirely by the retrieved 83%— $\approx +0.039$ F1 per retrieved sample.

Caveat. At $N=100$ the per-cell binomial noise on EM-style metrics is $\approx \pm 0.04$ (one s.d.), so per-category deltas of that size are directional; the base-model lift on English and the EN-vs-mixed contrast are the robust findings. The cells are paired on identical questions, so a paired test—not the marginal band—is the appropriate significance check. As a representativeness check, an independent $N=500$ run of English base_norag gives F1 0.690, matching the $N=100$ cell (0.688) within 0.003.

Conclusion of the ablation. On a *self-contained* benchmark, once the model is fine-tuned *on the test language*, retrieval adds no evidence the model lacks and only costs context tokens—substitutes. Retrieval retains value in the un-tuned model (format supply) and, directionally, where fine-tuning is weakest. This distinguishes closed-context temporal reasoning from open-domain temporal QA, where retrieval fills a genuine information gap.

VII. LIMITATIONS

Results come from a single seed and one model family, so conclusions are demonstrated for the 3B-QLoRA regime rather than proven in general. The ablation cells use $N=100$ ($N=25$ per language on mixed), supporting the aggregate contrasts but leaving fine-grained deltas directional. Translation quality is bounded by NLLB-200—residual noise on isolated entities depresses French most—and the *context-stuffing* RAG mode is implemented but not systematically swept. Finally, free-form tgqa answers are imperfectly served by string-overlap metrics.

VIII. REPRODUCIBILITY

The package is `multilingual_rag_tiser/`; datasets are git-LFS tracked (`git lfs install && git lfs pull`, otherwise a clone yields pointer files). Runner scripts cover training, evaluation, the RAG ablation, mixing, translation, and scoring. The frozen 15k snapshot and its provenance verifier (Section III) pin the English training set; result JSONs behind every table are committed under `experiments/`, the fine-tuned LoRA adapters are on the Hugging Face Hub (user `arianmo47`; links in the repository README), and unit tests (`python -m pytest tests/`) cover the samplers and the metrics.

IX. CONCLUSION

A 3B QLoRA model reaches F1 0.912 on English TISER within 8 GB (base: 0.690); one four-language fine-tune matches this on English while adding IT/DE/FR at F1 0.865–0.890 with $\text{EngLeak} \leq 0.008$, whereas an English-only fine-tune transfers only partially—the multilingual pipeline earns its cost. The RAG ablation shows retrieval and fine-tuning are *substitutes* on English (base +0.049, ft −0.024 F1), while on the mixed set—where fine-tuning is weakest—retrieval trends positive (+0.032 F1, directional). Retrieval’s value depends on whether a gap—informational or linguistic—remains that fine-tuning has not closed. Main challenges: NLLB noise on dates and entities (especially French), the 8 GB budget, and free-form tgqa answers.

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